

M.Tech Research Literature Survey

Project Title: AI-Based Digital Brain Twin for Predicting Neurological Disease Progression Using MRI, EEG and Clinical Data

1. Required Literature Survey Table

S.NO	TITLE OF THE PAPER	NAME OF JOURNAL / CONFERENCE	AUTHORS	VOLUME, ISSUE, YEAR	METHODOLOGY USED	APPLICATION	OBSERVATION (RESULTS, MERITS & DEMERITS)	DATASET SOURCE & DESCRIPTION	PAPER LINK	DATASET LINK
1	Tutorial: Brain Network Analysis with Graph Neural Networks	Not Reported	Hejie Cui, Xuan Kan, Xiaoxiao Li, Lifang He, Liang Zhan, Ying Guo, Carl Yang	Not Reported (Tutorial Paper)	Graph Neural Networks (GNNs), BrainGB framework space mapping, message-passing mechanism variants, attention enhancement paradigms, pooling selection strategies.	Comprehensive review of structural/functional connectome pipelines and modeling recommendations.	Results: Not Reported. Merits: Standardizes foundational GNN architectures into modular spaces; offers out-of-the-box BrainGB execution scripts. Demerits: Identifies heavy constraints due to ethical privacy hurdles, leading to thin cross-validation data.	Benchmarking repository framework built across multi-site connectome structures.	hejie.cui@emory.edu	https://braingb.us

S.NO	TITLE OF THE PAPER	NAME OF JOURNAL / CONFERENCE	AUTHORS	VOLUME, ISSUE, YEAR	METHODOLOGY USED	APPLICATION	OBSERVATION (RESULTS, MERITS & DEMERITS)	DATASET SOURCE & DESCRIPTION	PAPER LINK	DATASET LINK
2	A Unified Approach to Interpreting Model Predictions	Neural Information Processing Systems (NIPS)	Scott M. Lundberg, Su-In Lee	31st Conference, 2017	SHAP (SHapley Additive exPlanations) values formulation, additive feature attribution framework, Kernel SHAP (linear LIME optimization), Deep SHAP (DeepLIFT linearization connection).	Local local interpretable visual/textual explanation mapping for black-box neural networks.	Results: Kernel SHAP improves sample efficiency over Shapley sampling. Merits: Unifies six historic model attribution systems under mathematically strict consistency properties. Demerits: Exact calculation of the conditional features expectation introduces $O(2^M + M^3)$ complexity bounds.	MNIST handwritten database, 10-feature tree structures, and multi-class classification setups.	Not Reported	https://github.com/slundberg/shap

S.NO	TITLE OF THE PAPER	NAME OF JOURNAL / CONFERENCE	AUTHORS	VOLUME, ISSUE, YEAR	METHODOLOGY USED	APPLICATION	OBSERVATION (RESULTS, MERITS & DEMERITS)	DATASET SOURCE & DESCRIPTION	PAPER LINK	DATASET LINK
3	AI-Augmented Adaptive Digital Twin Modeling for Brain Tumor Evolution Prediction and Treatment Scheduling	Not Reported	Wenxi Liu, Michael Trimboli, Xianqi Li	Published 2026	Mechanistic reaction-diffusion (RD) spatial equations, logistic proliferation field, 3D residual block augmentation (U-Net structure), Model Predictive Control (MPC) look-ahead loops, recursive rollout updating.	Personalized spatiotemporal glioblastoma evolution forecasting and constrained chemotherapy/ radiotherapy regimen scheduling.	Results: Residual correction cuts voxel MSE by 84.3%, boosts Dice by 43.5%; online twin tracking drops rollout drift by 45.9%. Merits: Fuses robust biological physical constants seamlessly with flexible learned data compensation layers. Demerits: Trajectory alignment capabilities drop under highly sparse observation sequences without separate hyperparameter tuning.	UPENN-GBM cohort; 387 synthetic multi-modal longitudinal 120-step evolution frames simulated over a \$128 imes128\$ isotropic matrix grid.	arXiv:2607.13877v1	Not Provided

S.NO	TITLE OF THE PAPER	NAME OF JOURNAL / CONFERENCE	AUTHORS	VOLUME, ISSUE, YEAR	METHODOLOGY USED	APPLICATION	OBSERVATION (RESULTS, MERITS & DEMERITS)	DATASET SOURCE & DESCRIPTION	PAPER LINK	DATASET LINK
4	Attention U-Net: Learning Where to Look for the Pancreas	Medical Imaging with Deep Learning (MIDL)	Ozan Oktay, Jo Schlemper, Loic Le Folgoc, Matthew Lee, Mattias Heinrich, Kazunari Misawa, Kensaku Mori, Steven McDonagh, Nils Y Hammerla, Bernhard Kainz, Ben Glocker, Daniel Rueckert	1st Conference, 2018	Attention Gate (AG) layers, grid-based local feature soft-attention pooling, 3D fully convolutional network, Sorensen-Dice multi-class balanced loss, deep-supervision optimization.	Multi-class abdominal CT segmentation and automated target boundaries feature isolation.	Results: Attained \$0.840\backslash\pm0.087\$ Pancreas Dice Score on multi-label abdominal CT-150 validation splits. Merits: Automatically prunes noisy or irrelevant background gradient parameters on the fly without distinct cascaded pre-localization blocks. Demerits: Demands heavy GPU memory footprints, restricting standard resolution scaling options.	TCIA Pancreas CT-82 (82 scans) and localized multi-class abdominal CT-150 dataset (150 scans).	Not Reported	https://github.com/ozan-oktay/Attention-Gated-Networks
5	A Survey of Deep Learning for Alzheimer's Disease	Machine Learning and Knowledge Extraction	Qinghua Zhou, Jiaji Wang, Xiang Yu, Shuihua Wang, Yudong Zhang	Vol. 5, Issue 2, Pages 611-668, 2023	Systematic review of 2D/3D CNN feature pipelines, Recurrent Networks (LSTM/GRU segments), Graph Neural Networks (Spectral/Spatial GCN layers), Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs).	AD/MCI diagnosis tier classification, longitudinal cognitive conversion mapping, and surrogate biomarker forecasting.	Results: Binary model categorization setups show diagnostic benchmarks ranging from 70% up to 95% + accuracy. Merits: Eliminates structural manual feature-crafting steps over high-dimensional medical records. Demerits: Highlights severe information leakage trends and absent out-of-distribution independent checking cohorts.	ADNI, OASIS, AIBL, NACC, EDSD, ARWIBO, HABS, KLOSCAD, VITA open repositories.	https://doi.org/10.3390/make5020035	http://gaain.org/

S.NO	TITLE OF THE PAPER	NAME OF JOURNAL / CONFERENCE	AUTHORS	VOLUME, ISSUE, YEAR	METHODOLOGY USED	APPLICATION	OBSERVATION (RESULTS, MERITS & DEMERITS)	DATASET SOURCE & DESCRIPTION	PAPER LINK	DATASET LINK
6	Digital Twin Brain: a simulation and assimilation platform for whole human brain	Not Reported	Wenlian Lu, Longbin Zeng, Xin Du, Wenyong Zhang, Shitong Xiang, Huarui Wang, Jiexiang Wang, Mingda Ji, Yubo Hou, Minglong Wang, Yuhao Liu, Zhongyu Chen, Qibao Zheng, Ningsheng Xu, Jianfeng Feng	arXiv preprint, 2023	Leaky Integrate-and-Fire (LIF) neuronal architecture, hierarchical random graphs, hierarchical mesoscopic data assimilation (HMDA) solver, Balloon-Windkessel hemodynamic mapping.	Human-brain-scale spatiotemporal spiking network functional simulation and cognitive activity assimilation.	Results: Simulated 86B neurons and 47.8T synapses within 65s per biological second; attained visual rating projection $r=0.655$. Merits: Maps personalized connectome records (sMRI, DTI matrices, tracer PET metrics) into parallel cluster networks. Demerits: Extreme memory-access bounds and massive unbalanced communication traffic routes across computing processors.	Individualized multi-modal imaging indexes derived via structural sMRI parcellation, DTI fiber track counts, and tracer-level PET scans.	arXiv:2308.01241v1	Not Provided

S.NO	TITLE OF THE PAPER	NAME OF JOURNAL / CONFERENCE	AUTHORS	VOLUME, ISSUE, YEAR	METHODOLOGY USED	APPLICATION	OBSERVATION (RESULTS, MERITS & DEMERITS)	DATASET SOURCE & DESCRIPTION	PAPER LINK	DATASET LINK
7	Digital twin technologies in medicine: The innovations, barriers, and future directions	Intelligent Hospital	Gowtham Menon, Bhagyashri Malave, Mayuri Mhaske, Sakshi Parjane, Hrushikesh Mhaismale	Vol. 2, 2026	Internet of Medical Things (IoMT) continuous data streams, Cyber-Physical Systems (CPS) interface links, Edge/Cloud distributed pipelines, big data infrastructure workflows.	Continuous health monitoring loops, proactive medical care intervention, and healthcare facility system optimization.	Results: Real-time continuous biometric evaluation loops yield health decline prediction tags hitting up to 95.4% accuracy. Merits: Suppresses traditional trial-and-error healthcare practices with interactive, personalized digital modeling protocols. Demerits: Rigid multi-vendor system interoperability blocks and lack of universal clinical approval channels.	Multi-source cross-sectional clinical registers, patient EHR datasets, and remote wearable sensor tracking records.	https://doi.org/10.1016/j.inhs.2025.100043	Not Provided

S.NO	TITLE OF THE PAPER	NAME OF JOURNAL / CONFERENCE	AUTHORS	VOLUME, ISSUE, YEAR	METHODOLOGY USED	APPLICATION	OBSERVATION (RESULTS, MERITS & DEMERITS)	DATASET SOURCE & DESCRIPTION	PAPER LINK	DATASET LINK
8	Digital twins for health: a scoping review	npj Digital Medicine	Evangelia Katsoulakis, Qi Wang, Huanmei Wu, Leili Shahriyari, Richard Fletcher, Jinwei Liu, Luke Achenie, Hongfang Liu, Pamela Jackson, Ying Xiao, Tanveer Syeda-Mahmood, Richard Tuli, Jun Deng	Vol. 7, Article 77, 2024	System paradigm categorization (static vs. functional mirror vs. intelligent cognitive twin systems), in-silico trials (IST) design, multiscale biological modeling pipelines.	Healthcare delivery process orchestration, virtual clinical trial layout simulation, and personalized treatment mapping.	Results: Virtual imaging trial pipelines (VICTRE) with 2,986 synthetic patient twins successfully validate tomosynthesis tool changes. Merits: Accelerates drug discovery pipelines and therapeutic protocol assessment while reducing early physical human cohort safety exposures. Demerits: Demographic training data imbalances present severe risks of exacerbating socioeconomic healthcare divides.	VICTRE breast lesion databases, Cleveland Clinic Twin Health diabetes tracking trials, and public healthcare registers.	https://doi.org/10.1038/s41746-024-01073-0	http://dt4h.org/

S.NO	TITLE OF THE PAPER	NAME OF JOURNAL / CONFERENCE	AUTHORS	VOLUME, ISSUE, YEAR	METHODOLOGY USED	APPLICATION	OBSERVATION (RESULTS, MERITS & DEMERITS)	DATASET SOURCE & DESCRIPTION	PAPER LINK	DATASET LINK
9	Digital Twins in Health Care: Ethical Implications of an Emerging Engineering Paradigm	Frontiers in Genetics	Koen Bruynseels, Filippo Santoni de Sio, Jeroen van den Hoven	Vol. 9, Article 31, 2018	Value-sensitive design engineering framework, philosophical conceptual analysis, multi-omics molecular profiling, personalized baseline tracking constraints.	Ethical governance mapping, critical evaluation of healthcare data-driven definitions, and anti-aging lifespan modeling analysis.	Results: Not Reported (Theoretical Analysis). Merits: Moves clinical definitions of normality from standard population distributions to individualized baseline patterns. Demerits: Underlying statistical risk scores present severe hidden discrimination vulnerabilities or loss of autonomy to data algorithms.	Conceptual multi-omics molecular and behavioral data-tracking paradigms.	doi: 10.3389/fgene.2018.00031	Not Provided
10	Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization	International Journal of Computer Vision	Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, Dhruv Batra	arXiv preprint 2016 / IJCV 2019	Gradient-weighted Class Activation Mapping (Grad-CAM), Global Average Pooling gradients tracking, Guided Backpropagation integration, Guided Grad-CAM visual fusion mapping.	Generating high-resolution class-discriminative visual and textual explanation maps for deep neural network choices.	Results: Enhances weakly-supervised localization over CAM while retaining strict model accuracy thresholds. Merits: Seamlessly compatible with diverse off-the-shelf CNN families (VGG, ResNet) without network alterations or re-training steps. Demerits: Heatmap resolution grids grow progressively coarser when generated within shallower layers.	ImageNet ILSVRC-15 validation matrices, PASCAL VOC 2007/2012 object sets, MSCOCO captions, Places365.	https://github.com/ramprs/grad-cam/	http://gradcam.cloudcv.org

2. Comparative Analysis Matrix

PAPER	STRENGTH	LIMITATION	RESEARCH GAP	HOW MY PROJECT IMPROVES IT
Cui et al. (BrainGB)	Standardizes structural/functional connectome architectures via the unified message-passing benchmark.	Constrained strictly to static, cross-sectional graph representations of node topologies.	No integration of multi-modal, highly non-linear parameters or continuous clinical logs over time.	Unifies continuous multi-modal grids: Fuses spatial structural dimensions (\$MRI\$) with rapid temporal electrical signals (\$EEG\$) and longitudinal clinical vectors.
Lundberg et al. (SHAP)	Establishes game-theoretic consistency validation proofs for local feature significance mappings.	Massive combinatorial tracking overheads prevent real-time 3D volumetric integration.	Lacks any built-in spatiotemporal modeling constants or active predictive loop parameters.	Deploys scalable 3D explainability: Integrates fast spatial attention gating blocks to map local volumetric parameters without exponential calculation limits.
Liu et al. (Tumor DT)	Augments classical mechanistic spatial cell reaction equations with data-driven residual U-Net adjustment fields.	Demands continuous, exceptionally high-frequency observation sequences to prevent rollout divergence.	Fails to incorporate real-time electrophysiological vectors or continuous cognitive parameters.	Stabilizes sparse multi-modal rollouts: Pairs spatial cell transport layers with high-resolution temporal \$EEG\$ modules under robust rolling adaptation states.
Oktay et al. (Attention U-Net)	Grid-based soft soft-attention pooling dynamically prunes background noise without decoupled networks.	Optimized strictly for structural target segmentation tasks; completely missing time-series tracking.	Fails to address chronic progression or long-term continuous parameter updating loops.	Maintains spatiotemporal continuity: Links 3D spatial feature maps with continuous recurrent loops to map multi-scale variables simultaneously.
Zhou et al. (AD Survey)	Delivers an exhaustive analysis across deep architectures, cataloging classification performance.	Identifies pervasive methodology flaws, data leakage issues, and the near-total absence of out-of-distribution validations.	Lacks a unified computational framework capable of actively updating parameters during long rollouts.	Implements rigorous data integrity: Utilizes strict patient-level cohort splits and a validated twin adaptation buffer to fully prevent information leakage.
Lu et al. (Digital Twin Brain)	Achieves rapid human-scale simulation containing up to 86 billion spiking biological neurons.	Requires massive compute resources; suffers from severe memory access unbalances across cluster links.	Focused entirely on simulating resting-state patterns without an active clinical optimization layer.	Targeted decision optimization: Focuses the virtual replica on localized region sets, coupling predictions with active Model Predictive Control (MPC).
Menon et al. (DT Technologies)	Provides a clear blueprint mapping cloud/edge infrastructure interactions for IoMT data streams.	Remains entirely conceptual, missing quantitative model benchmarks or direct dataset validations.	A major gap persists between theoretical architectural definitions and exact biological simulation models.	Delivers an executable workflow: Provides a concrete pipeline mapping spatial frame tracking data directly to constrained active predictive loops.

PAPER	STRENGTH	LIMITATION	RESEARCH GAP	HOW MY PROJECT IMPROVES IT
Katsoulakis et al. (Scoping Review)	Provides crisp, precise digital twin definitions and assesses real-world validation frameworks like VICTRE.	Clinical targets are siloed across standalone processes or single isolated organs.	No unified multi-modal mathematical framework exists to combine spatial structures with continuous brain signals.	Fuses structural and electrical signatures: Integrates multi-modal dependencies within a single co-calibrated virtual mapping matrix.
Bruynseels et al. (Ethical Paradigm)	Establishes that personalized tracking benchmarks can successfully relocate clinical normality parameters.	Lacks any technical code architectures or verifiable data processing frameworks.	Missing a computational engine to test and observe theoretical personal tracking limits under active control.	Translates values to control routines: Embeds personalized baseline matching equations inside a constrained MPC scheduling block.
Selvaraju et al. (Grad-CAM)	Generates off-the-shelf class-discriminative localization cues without network retraining steps.	Resolution details grow progressively coarser within shallower network layer steps.	Strictly confined to passive explanation check tasks; missing active treatment management components.	Links explainability to control: Pairs visual class tracking indices directly with look-ahead MPC routines to optimize clinical scheduling.

3. Final Synthesis & Strategic Project Plan

3.1 Common Methodologies & Dataset Paradigms

The state-of-the-art literature in deep medical imaging relies heavily on 3D Convolutional Neural Networks (3D-CNNs) and U-Net-based variants to extract spatial features from volumetric records. To address chronic conditions, these architectures are increasingly paired with Gated Recurrent Units (GRUs) or Long Short-Term Memory (LSTM) systems to capture continuous temporal trends. For network-level structural mapping, Graph Convolutional Networks (GCNs) serve as the standard. These networks are evaluated using metrics like Accuracy, Area Under the Curve (AUC-ROC), and Dice Similarity Coefficients (DSC). Data sourcing is predominantly rooted in historical open-access cohorts such as ADNI, OASIS, and the UPENN-GBM cohort.

3.2 Critical Literature Limitations & Unexplored Research Gaps

The literature analysis reveals several systemic vulnerabilities:

- **Exposure Bias and Rollout Drift:** Models optimized for short-term prediction accumulate intense voxel-wise density errors during recursive long-horizon rollouts, leading to data degradation.
- **Data Leakage and Validation Failures:** A high percentage of publications fail to implement patient-level validation splits, cross-contaminating training and testing structures.
- **Siloed Modality Processing:** Current paradigms process high-resolution structural records (\$MRI\$) and dynamic electrophysiological signals (\$EEG\$) as independent vectors, missing critical spatiotemporal interactions.
- **Passive Prediction vs. Active Control:** Extant frameworks operate strictly as passive estimators without linking predictions to active, look-ahead treatment optimization engines.

3.3 Methodological Implementation: How My Project Overcomes These Limitations

Your proposed project, "**AI-Based Digital Brain Twin for Predicting Neurological Disease Progression Using MRI, EEG and Clinical Data,**" directly bridges these gaps through a unified prediction-decision pipeline:

- **Unified Spatiotemporal Fusion Framework:** Instead of processing modalities in isolation, your architecture builds a multi-channel input framework where 3D spatial grids (\$MRI\$), continuous temporal signals (\$EEG\$), and clinical vectors are co-calibrated within a single learning matrix.
- **AI-Augmented Hybrid Mechanistic Core:** By anchoring the framework to a fixed reaction-diffusion mathematical core, you retain strict physical properties (tissue-dependent diffusion and carrying capacity constraints). The 3D residual block (U-Net layer) focuses exclusively on learning complex, patient-specific model-form discrepancies.
- **Rollout Stabilization via Online Adaptation:** To eliminate rollout drift and exposure bias, your framework incorporates an explicit short-window patient adaptation buffer. This updates and recalibrates the residual network parameters on the fly during deployment, preserving long-term validation accuracy.
- **Closed-Loop Model Predictive Control (MPC) Layer:** Your project transforms passive estimation into an active, decision-oriented system. The updated digital twin functions as the local predictor inside an MPC shooting layer, optimizing candidate chemotherapy and radiotherapy schedules over a look-ahead planning horizon under real-world toxicity, dose, and uncertainty bounds.